

Project Executable Files

Date	03 July 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction System
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example or similar)	NA
6	Deployed application URL (if hosted)	Yes
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	Yes
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

Overview of the project's file and folder structure as submitted:

```

credit-card-approval-prediction/
  app.py          - Flask application entry point (loads model, serves /predict)
  credit_card_model.pkl - Trained Random Forest model (serialized with joblib)
  CreditCard.ipynb  - Model training, EDA and evaluation notebook
  creditcard.csv    - Transaction dataset used to train/evaluate the model
  templates/
    index.html     - Landing page / approval prediction UI (Tailwind-based)
  static/
    style.css      - Stylesheet (currently not linked from index.html)

```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Not deployed – currently runs locally using Flask (http://127.0.0.1:5000)
Login Credentials (Demo)	Not Applicable
Platform / Hosting Provider	NA - not yet deployed to a hosting provider
Repository Link	https://github.com/aliasuhana2005-lab/Credit-Card-Approval-Prediction-System
Demo Video Link	https://drive.google.com/file/d/1HbxAqPBaK1lP2SSs1lG0tKWldwDsQDxp/view

Step 4: Run Instructions

Step-by-step instructions for the evaluator to run the project locally:

Pre-requisites:

Python 3.9+ and pip. Required packages: flask, numpy, joblib, scikit-learn (the model was trained and pickled with scikit-learn 1.9.0 - installing the same version avoids compatibility warnings when unpickling).

Step 1: Place app.py, credit_card_model.pkl, and the templates/ folder (containing index.html) in the same project directory.

Step 2: Install dependencies: `pip install flask numpy joblib scikit-learn`

Step 3: Add the missing predict.html and result.html templates referenced by app.py (see Known Issues below) - the app will not run end-to-end without them.

Step 4: Start the application: `python app.py`

Step 5: Open a browser at <http://127.0.0.1:5000/predict> to submit applicant/transaction details and view the prediction result.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	app.py renders predict.html and result.html, but only index.html was submitted	Add the two missing templates before the app can run end-to-end

S.No	Known Issue / Limitation	Workaround / Status
2	No requirements.txt / dependency file included	Install flask, numpy, joblib and scikit-learn manually as listed in Step 4
3	style.css is included but not linked from index.html (index.html uses Tailwind via CDN instead)	Either link style.css in a template that needs it, or remove it if unused
4	App is served via Flask's single-threaded development server (app.run(debug=True)), which the Performance Testing report showed exceeds the 80% CPU target under concurrent load	Deploy behind a production WSGI server (e.g. Gunicorn) before hosting publicly